CDI COLLEGE

ASSIGNMENT 2

Technical Analysis Using R

Preliminary Stage

Student Name	Sara Mohamed Mohamed Flefla
Course	BDA400 – Data Science Tools and Techniques
Assignment	CA_BDA400_A02
Portfolio Symbols	AMZN, GOOGL, TSLA
Submission Date	November 12, 2026

GitHub Repository

https://github.com/sarafleflaca-hub/Technical-analysis-sara

This report summarizes the R workflow, portfolio analysis, generated outputs, and files uploaded to GitHub.

1. Introduction and Objectives

This report documents the preliminary stage of a technical-analysis project using R. The work covers the RStudio environment, required packages, portfolio configuration, stock-data importing, descriptive statistics, and visual outputs.

Project objectives

• Configure and use RStudio.
• Set up a portfolio containing AMZN, GOOGL, and TSLA.
• Import stock time-series data and calculate summary statistics.
• Generate closing-price charts and export a CSV summary.
• Organize the deliverables and upload them to GitHub.

2. GitHub Account and Repository

The project is available in Sara's public GitHub repository: Technical-analysis-sara. The repository screenshot confirms the main project files at the repository root and a separate **Stock_Charts** folder.

Files observed: Flefla_Sara_CA_BDA400_A02.docx; Flefla_Sara_CA_BDA400_A02.pdf; README.txt; portfolio.txt; stock_statistics.csv; technical_analysis.R; and Stock_Charts/ containing AMZN.png, GOOGL.png, and TSLA.png.

Folder-structure note: The current GitHub screenshot shows project files in the repository root, not inside an Assignment2 subfolder. If the instructor specifically requires an Assignment2 subfolder, move or copy the deliverables there before submission.

3. R and RStudio Environment

RStudio was opened successfully on Windows. The captured Console identifies R version 4.6.2 and shows the working directory as C:/Users/ADMIN/Downloads/Flefla_Sara_CA_BDA400_A02_package. Installation evidence is included later in this report.

4. Required R Packages

The project script checks for and uses the following packages:

Package	Purpose
quantmod	Retrieving market data for selected stock symbols.
TTR	Technical-analysis calculations, including moving averages.
xts	Working with time-indexed financial data.
ggplot2	Creating closing-price charts.

5. Portfolio Configuration and Data Import

The portfolio.txt file contains one ticker symbol per line: AMZN, GOOGL, and TSLA. The script reads these symbols, imports the corresponding stock time-series data, and stores the successfully imported objects in portfolio_data. The R Console confirmed that names(portfolio_data) returned AMZN, GOOGL, and TSLA.

6. Summary Statistics

The recorded analysis calculates the latest 20-day simple moving average, mean, mode, median, and standard deviation using closing prices. The following values are reproduced from the captured RStudio output:

Symbol	20-day MA	Mean	Mode	Median	Std. Dev.
AMZN	255.8675	241.9198	222.69	244.85	19.90911
GOOGL	308.1835	333.8298	318.58	336.02	28.88584
TSLA	381.8350	395.0312	372.11	395.23	38.02328

These figures reflect the recorded run and may change when the script is run again because market prices and available data can change.

7. Data Display and Visualizations

The script generates closing-price line charts for all three portfolio symbols and saves the PNG files in the Stock_Charts folder. The RStudio evidence below shows the TSLA Closing Price chart. The GitHub folder screenshot confirms that AMZN.png, GOOGL.png, and TSLA.png were uploaded.

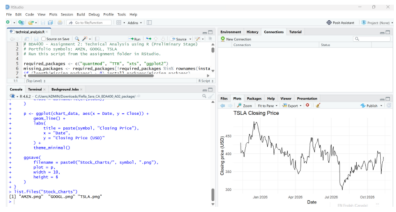


Figure 1. RStudio confirms AMZN.png, GOOGL.png, and TSLA.png; TSLA closing-price chart is displayed.

The script also exports the combined summary statistics to stock_statistics.csv. Confirm that the CSV file is present in the project folder and uploaded to GitHub.

9. How to Run the Project

1. Open the project folder in RStudio. 2. Confirm portfolio.txt and technical_analysis.R are in the working directory. 3. Run the script from top to bottom using Source. 4. Allow missing packages to install. 5. Review the Console for data and statistics. 6. Confirm the three PNG files in Stock_Charts. 7. Confirm stock_statistics.csv exists. 8. Upload all deliverables to Sara's TechnicalAnalysis/Assignment2 folder.

10. Project Deliverables

portfolio.txt; technical_analysis.R; stock_statistics.csv; Stock_Charts/AMZN.png; Stock_Charts/GOOGL.png; Stock_Charts/TSLA.png; README.txt or README.md; and this report in DOCX/PDF format.

11. Conclusion

The R workflow for Sara's portfolio has been executed: AMZN, GOOGL, and TSLA were loaded into portfolio_data, summary statistics were calculated, and three closing-price chart files were generated. The repository URL and remaining setup screenshots must be added before final submission.

12. Final Submission Checklist

Repository URL: Add Sara's confirmed TechnicalAnalysis repository link here.

Requirement	Evidence / Status
GitHub repository and URL	Needs confirmation / screenshot
Assignment2 uploaded	Needs GitHub screenshot
R and RStudio	RStudio running; installation evidence still needed
Required packages	Script uses/checks packages; add Console evidence
portfolio.txt	AMZN, GOOGL, TSLA
technical_analysis.R	Present in working folder
stock_statistics.csv	Confirm file exists
Three chart images	Confirmed: AMZN.png, GOOGL.png, TSLA.png

Sara Mohamed Mohamed Flefla • BDA400 Assignment 2 • Page 3

Figure 1. RStudio Console output and TSLA closing-price chart.

8. Exported Statistics

The script exports combined summary statistics to stock_statistics.csv. The original local project-folder evidence confirms that this CSV exists, and the GitHub repository screenshot confirms that the file is present in the repository.

9. How to Run the Project

1. Open the project folder in RStudio.
2. Confirm portfolio.txt and technical_analysis.R are in the working directory.
3. Run the script from top to bottom using Source.
4. Allow required packages to install if they are missing.
5. Review the Console for imported data and summary statistics.
6. Confirm the three PNG files in Stock_Charts.
7. Confirm stock_statistics.csv exists.

10. GitHub Upload Evidence

The screenshot below shows the public Technical-analysis-sara repository and confirms that the report files, R script, portfolio file, README, CSV summary, and Stock_Charts folder are present.

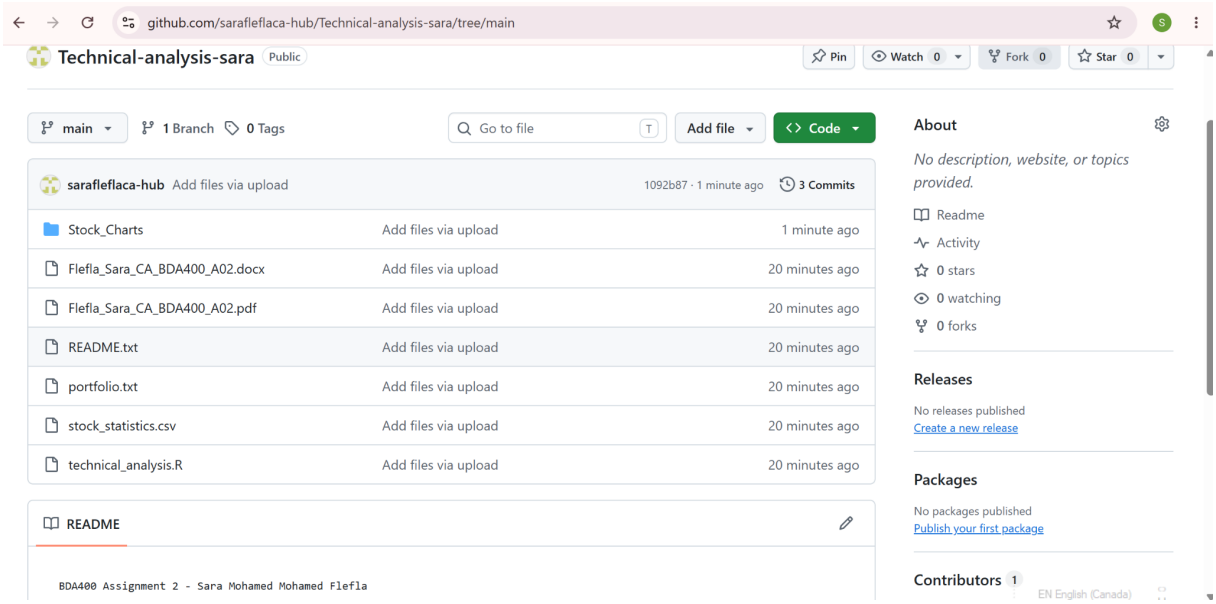


Figure 2. GitHub repository main page showing the uploaded project files.

The next screenshot shows the Stock_Charts folder and confirms the three chart image files.

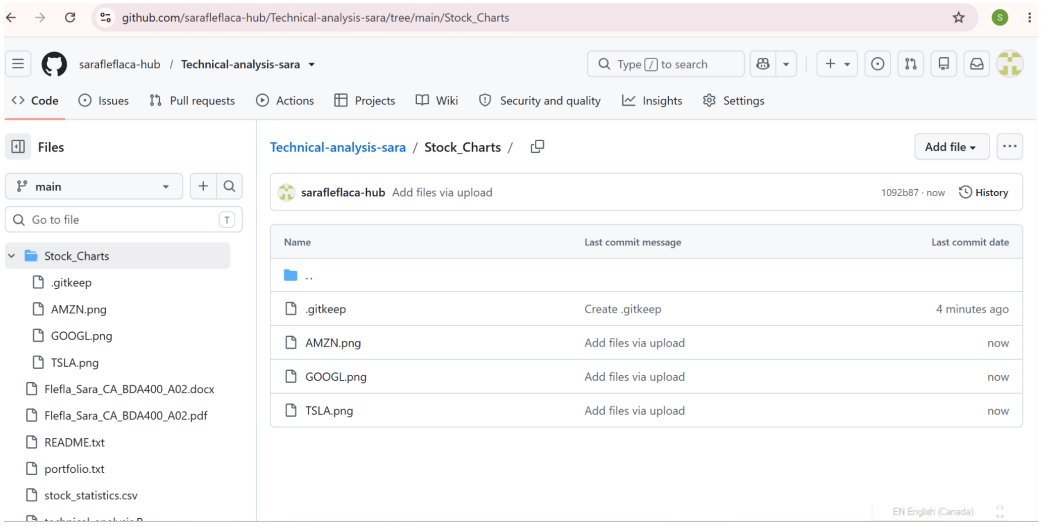


Figure 3. GitHub Stock_Charts folder showing AMZN.png, GOOGL.png, and TSLA.png.

11. RStudio Installation Evidence

The original report includes screenshots of the RStudio setup process. They document the setup welcome screen, installation location, installation in progress, and successful completion.

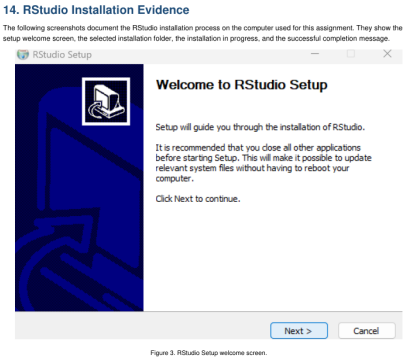


Figure 4. RStudio setup welcome screen.

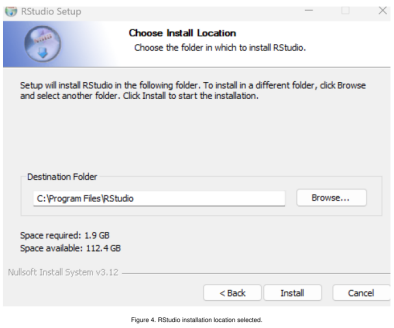


Figure 5. RStudio installation location selected.

12. RStudio Installation Completion

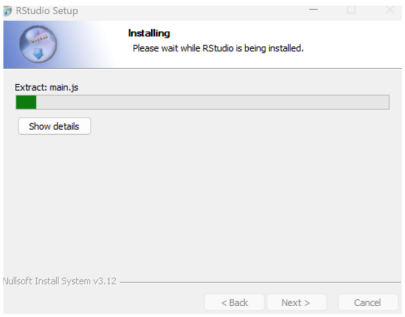


Figure 5. RStudio installation in progress.

Figure 6. RStudio installation in progress.

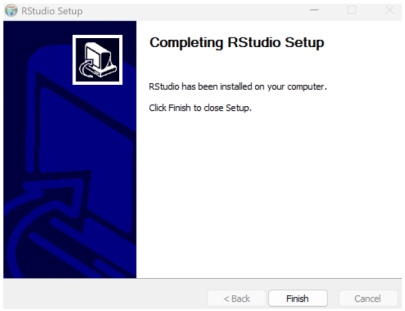


Figure 6. RStudio installation completed successfully.

Evidence note: The final setup screen states that RStudio has been installed on the computer. These screenshots establish installation only; they do not independently verify GitHub upload or assignment submission.

Figure 7. RStudio setup completed successfully.

The final setup screen states that RStudio has been installed on the computer. These screenshots establish installation only; they do not independently verify course-platform submission.

13. Final Submission Checklist

Requirement	Current evidence/status
GitHub repository	Public repository exists; link included in this report.
PDF and Word reports	Both files are present in the repository root.
R script	technical_analysis.R is present in the repository root.
Portfolio symbols	portfolio.txt contains AMZN, GOOGL, and TSLA.
Summary CSV	stock_statistics.csv is present in the repository root.
Chart images	AMZN.png, GOOGL.png, and TSLA.png are present in Stock_Charts/.
README	README.txt is present in the repository root.
Assignment2 subfolder	Not shown in the current repository screenshot; check whether the instructor requires this exact folder structure.
Course-platform submission	Must be completed separately in the course LMS; GitHub upload alone does not confirm submission.

14. Conclusion

The preliminary R workflow for Sara's portfolio has been executed. The project imported AMZN, GOOGL, and TSLA into portfolio_data, calculated summary statistics, generated three closing-price chart files, and exported a CSV summary. The GitHub screenshots confirm that the core deliverables and chart images have been uploaded.

Final reminder: verify the instructor's required repository naming and folder structure, and submit the PDF through the course LMS. The presence of files on GitHub does not by itself confirm LMS submission.

GitHub Repository: https://github.com/sarafleflaca-hub/Technical-analysis-sara